Netaji Subhas University of Technology

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Hardware and Software Tools Workshop

Subject Code: COCSC603

Retail Sales Data Visualization and Analytics using R

Project Report

Submitted by Team

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Project Repository: https://github.com/dheeraj-k-310/R-Library

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Abstract

Data visualization plays an important role in understanding large datasets and identifying meaningful insights. This project focuses on analyzing retail sales data using the R programming language and its powerful visualization libraries. The project performs data cleaning, transformation, and exploratory analysis to identify sales trends, top-performing products, and regional sales distribution.

Using libraries such as ggplot2, dplyr, tidyr, readr, and lubridate, the raw sales dataset is processed and visualized through different charts and graphs. The analyzed data can further be exported for interactive dashboards in Power BI. The goal of this project is to demonstrate how R can be used effectively for data analytics and visualization in business decision making.

Problem Statement

Retail businesses generate large amounts of sales data every day. However, raw datasets are often difficult to interpret and do not provide clear insights for decision making.

Businesses need to understand:

- Sales trends over time
- Top selling products
- Category wise revenue
- Regional sales distribution

Without proper tools and analysis, it becomes difficult to extract meaningful insights from raw sales data. Therefore, a system is required that can clean, analyze, and visualize retail sales data efficiently using R programming.

Objectives

The main objectives of this project are:

- To analyze retail sales data using the R programming language.
- To perform data preprocessing and cleaning.
- To identify sales trends over time.
- To analyze region-wise and category-wise sales performance.
- To identify top-selling products.
- To generate data visualizations using ggplot2.
- To prepare processed data for further visualization tools such as Power BI.

Tools and Technologies Used

Tool / Technology	Purpose
R Programming	Data analysis and visualization
ggplot2	Creating charts and graphs

Tool / Technology	Purpose
dplyr	Data manipulation
tidyr	Data cleaning
readr	Reading CSV datasets
lubridate	Handling date and time
Power BI	Dashboard visualization
CSV Dataset	Retail sales dataset

Dataset Description

The dataset used in this project contains retail sales information including product details, categories, regions, and revenue generated from each sale.

The dataset includes the following attributes:

- Order ID – Unique identifier for each order
- Product Name – Name of the product sold
- Category – Category of the product
- Region – Sales region where the product was sold
- Sales Amount – Revenue generated from the sale
- Order Date – Date when the order was placed

The dataset is stored in CSV format and imported into R for analysis.

System Architecture

The system architecture describes the workflow of the project from data collection to visualization.

Step 1 Data Collection — Retail sales data is stored in a CSV file.

Step 2 Data Loading — The dataset is imported into R using the readr library.

Step 3 Data Cleaning — The data is cleaned using dplyr and tidyr to remove missing values and format the dataset properly.

Step 4 Data Analysis — Exploratory Data Analysis (EDA) is performed to identify patterns and trends.

Step 5 Data Visualization — Graphs and charts are created using the ggplot2 library.

Step 6 Data Export — Processed data is exported as a CSV file (aapl_stock_data.csv) which can be imported directly into Power BI. Using Power BI's "Get Data" feature, this CSV can be loaded to build interactive dashboards with slicers, drill-through filters, and dynamic charts — enabling stakeholders to explore metrics such as closing price, RSI, moving averages, and monthly returns without requiring R.

Retail Sales Dataset (CSV) → Data Loading in R → Data Cleaning → Data Analysis → Visualization (ggplot2) → Power BI

Methodology

The project follows a structured methodology for analyzing the dataset.

1. Data Import

The dataset is imported into R using the read_csv() function.

```
sales_data <- read_csv("sales_data.csv")
```

2. Data Cleaning

The dataset is cleaned by removing missing values and formatting the columns correctly.

```
sales_data <- sales_data %>%
  drop_na()
```

3. Data Transformation

Data is organized and grouped using dplyr.

```
sales_summary <- sales_data %>%
  group_by(Category) %>%
  summarise(TotalSales = sum(Sales))
```

4. Exploratory Data Analysis

The dataset is analyzed to understand product performance, category performance, sales distribution, and sales trends over time.

5. Data Visualization

Charts and graphs are generated using ggplot2.

```
ggplot(sales_summary, aes(x = Category, y = TotalSales)) +
  geom_bar(stat = "identity") +
  theme_minimal()
```

Data Visualization Techniques

The project includes multiple visualization techniques to represent the data effectively.

1. Sales Trend Line Chart

Shows how sales change over time, enabling identification of seasonal patterns and growth periods.

2. Category-wise Sales Bar Chart

Displays the revenue generated by each product category for comparative analysis.

3. Region-wise Sales Chart

Shows how sales are distributed across different geographical regions.

4. Top Products Chart

Identifies the best performing products based on cumulative sales figures.

These visualizations help understand the dataset in a more intuitive and meaningful way, supporting informed business decisions.

Advantages of the System

- Provides clear insights from raw data
- Helps businesses identify top performing products
- Improves decision making using data-driven approaches
- Enables visual representation of large datasets
- Seamlessly integrates with Power BI — the exported CSV can be loaded into Power BI Desktop via "Get Data → Text/CSV" to build interactive dashboards with date slicers, KPI cards for close price and RSI, and dynamic line charts, making insights accessible to non-technical business users

Limitations

- The dataset used is a sample dataset; not real production data
- Real-time data integration is not currently implemented
- Advanced predictive analytics are not included in this version
- The analysis is limited to the available dataset attributes

Future Enhancements

The project can be improved in the future by incorporating the following:

- Adding machine learning models for sales prediction
- Integrating real-time retail datasets via APIs
- Creating interactive dashboards using R Shiny
- Adding customer behavior and sentiment analysis
- Implementing full predictive analytics pipeline

Conclusion

This project demonstrates how the R programming language can be effectively used for retail sales data analysis and visualization. By using libraries such as ggplot2, dplyr, tidyr, readr, and lubridate, the raw dataset is cleaned, analyzed, and transformed into meaningful visual insights.

The project helps identify sales trends, top performing products, and regional performance — all of which are critical for strategic business decision making. Furthermore, the processed data exported as CSV can be integrated with Power BI dashboards: by connecting the output file through "Get Data → Text/CSV" in Power BI Desktop, users can create interactive visuals such as date-range slicers, KPI summary cards, and dynamic trend charts — enabling non-technical stakeholders to explore the same insights without requiring R or any coding environment.

Overall, this project highlights the importance of data analytics and visualization in modern business environments and demonstrates the practical capabilities of R as an analytics platform.

Smart Stock Market Analysis using R

This section documents a separate B.Tech CSE mini-project for stock market analysis and visualization using R. The project downloads real-time stock data, performs technical analysis, and generates a comprehensive dashboard of charts and indicators.

Project Features

- Download stock data from Yahoo Finance
- Closing price trend analysis and daily return calculation
- Moving average analysis (50-day and 200-day MA)
- Volume analysis and RSI (Relative Strength Index) indicator
- Bollinger Bands computation and visualization
- 30-day ARIMA forecast for stock price prediction
- Multi-stock correlation analysis (AAPL, MSFT, GOOG)
- Risk analysis and dashboard-style chart output

Required R Packages

Install the following packages once in R before running the script:

```
install.packages(c(
  "quantmod", "ggplot2", "dplyr", "tidyquant",
  "forecast", "TTR", "corrplot"
))
```

How to Run

Open R or RStudio, navigate to the project folder, and execute the following command:

```
setwd("path/to/smart-stock-market-analysis-r")
source("stock_analysis.R")
```

Notes: The default stock ticker is AAPL with data starting from 2022-01-01. Internet access is required to download Yahoo Finance data.

Execution Steps

The script executes the following pipeline when run in RStudio:

> **Step 1** Package Installation — Install required R packages: quantmod, ggplot2, dplyr, tidyquant, forecast, TTR, corrplot.
>
> **Step 2** Set Working Directory — Use setwd() to navigate to the project folder in RStudio.
>
> **Step 3** Data Download — The script fetches AAPL stock data from Yahoo Finance using the quantmod library from 2022-01-01 onward.
>
> **Step 4** Technical Indicators — Computes MA50, MA200, RSI, Bollinger Bands, MACD, 20-day volatility, and drawdown metrics.
>
> **Step 5** Multi-stock Correlation — Analyzes correlation between AAPL, MSFT, and GOOG stock returns using the corrplot library.

Step 6 ARIMA Forecast — Generates a 30-day forward price forecast using the forecast library.

Step 7 Save Outputs — Stock data is exported as CSV to stick_data_csv/ and all plots are saved to stock_plots/ directory.

Yahoo Finance → quantmod Download → Technical Indicators → ARIMA Forecast → Plots & CSV Export

Script Execution Output

The following output was produced when stock_analysis.R was executed in RStudio on March 17, 2026. The analysis uses AAPL stock data from 2022-01-01 to the present date.

Figure 1: Technical Chart — Candlestick with MA50, MA200, and RSI (AAPL, 2022–2026)

Latest Market Snapshot (AAPL)

```
Date: 2026-03-17
Close Price: 254.23
Daily Return: 0.56 %
RSI: 40.78
20 Day Volatility: 1.33 %
Current Drawdown: -11.17 %
```

Figure 2: Apple Stock Closing Price (2022–2026)

Additional Performance Metrics

```
Average Daily Return:   0.0476 %
Best Daily Return:      15.33 %
Worst Daily Return:    -9.25 %
Maximum Drawdown:      -33.43 %
Avg 20 Day Volatility:  1.65 %
```

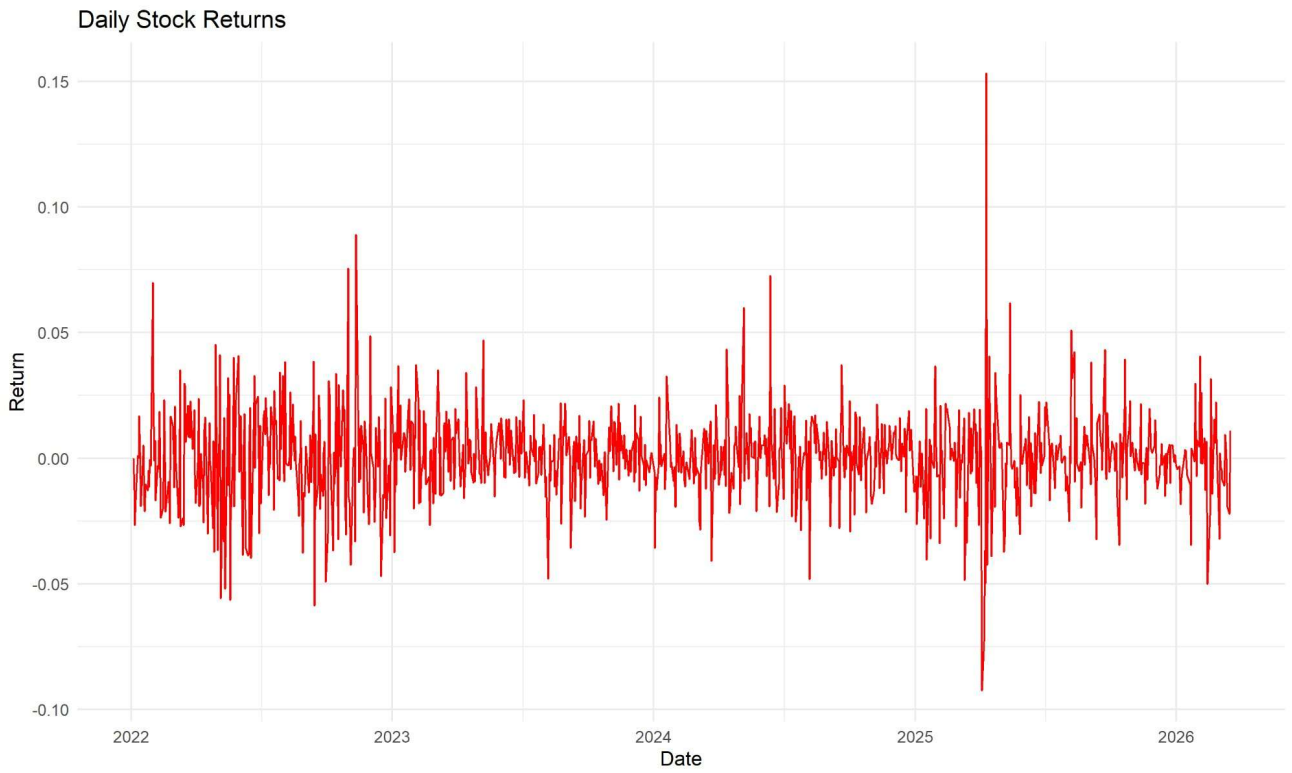

Figure 3: Daily Stock Returns (2022–2026)

Recent Monthly Returns

```
month      monthly_return
2025-10         +5.84 %
2025-11         +3.64 %
2025-12         -3.97 %
2026-01         -4.25 %
2026-02         -2.16 %
2026-03         -3.96 %
```

Figure 4: Moving Average Analysis — MA50 (blue) and MA200 (red)

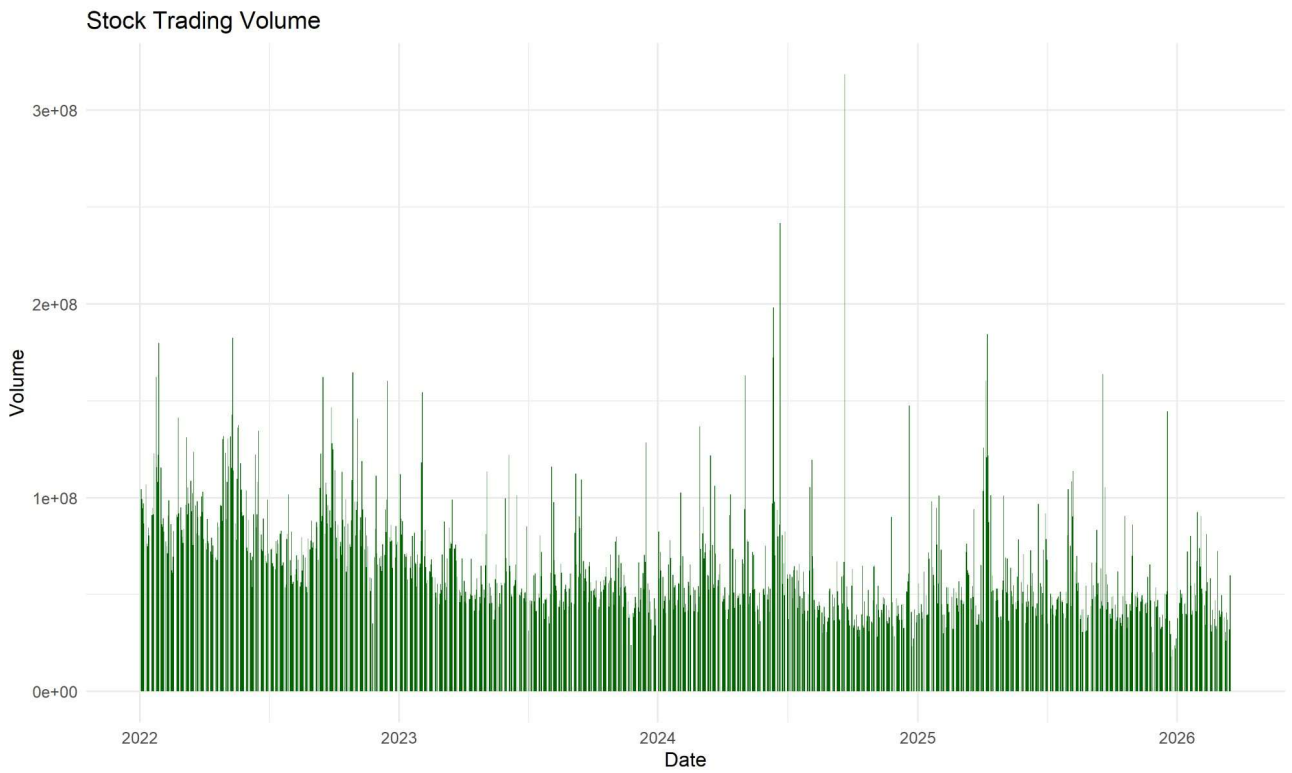

Figure 5: Stock Trading Volume (2022–2026)

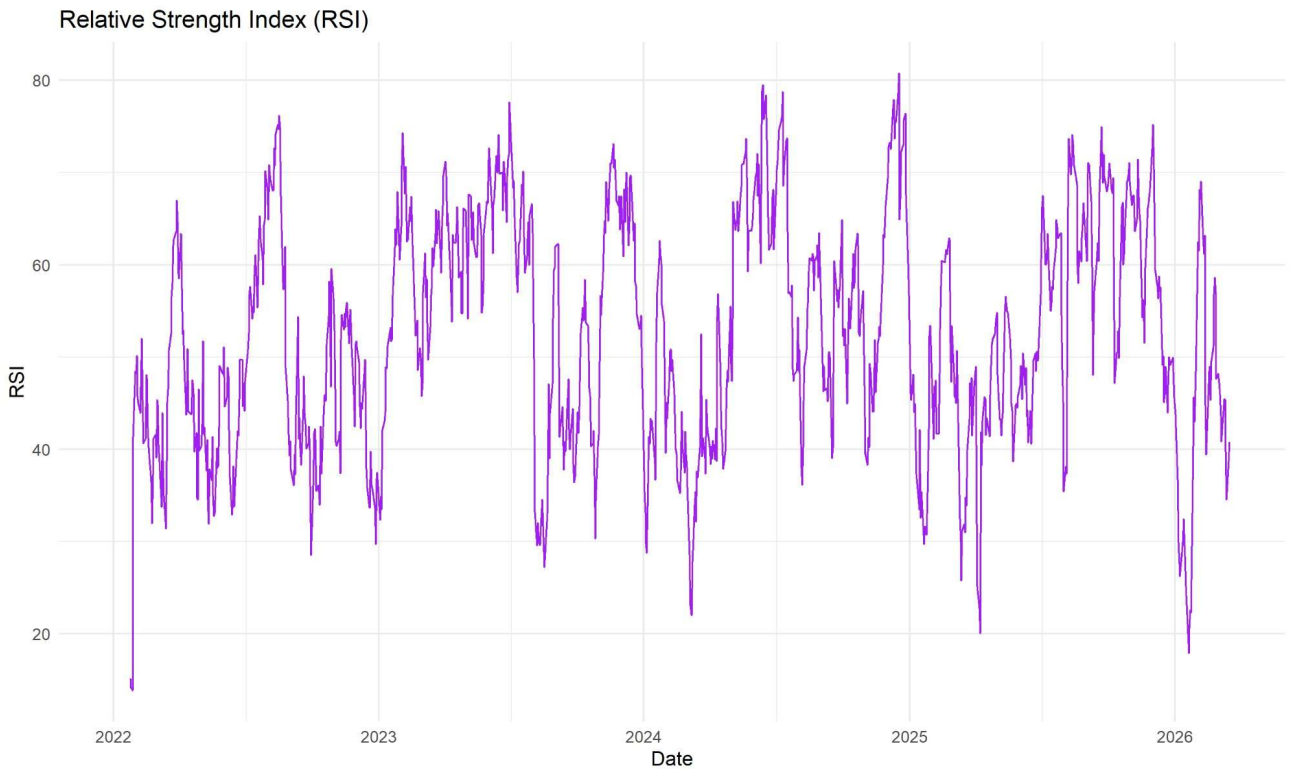

Figure 6: Relative Strength Index — RSI(14)

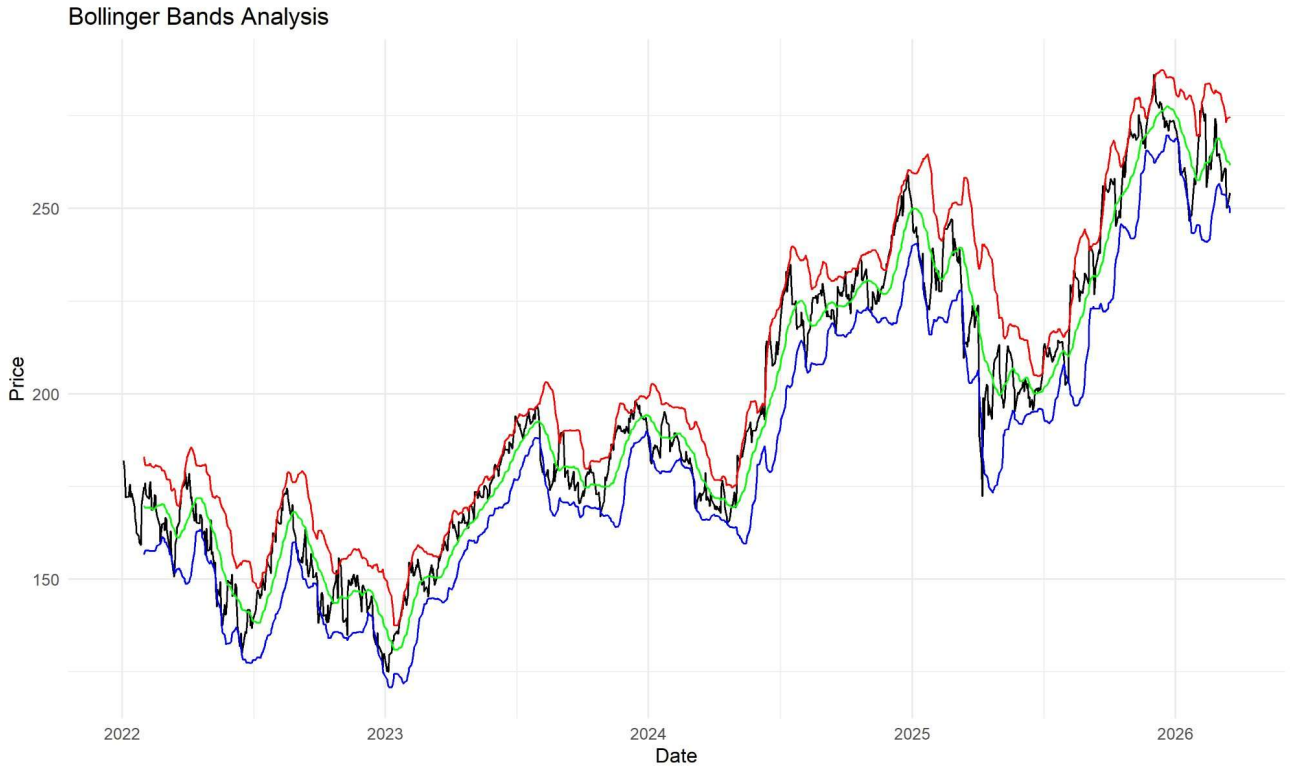

Figure 7: Bollinger Bands Analysis (Upper, Middle, Lower)

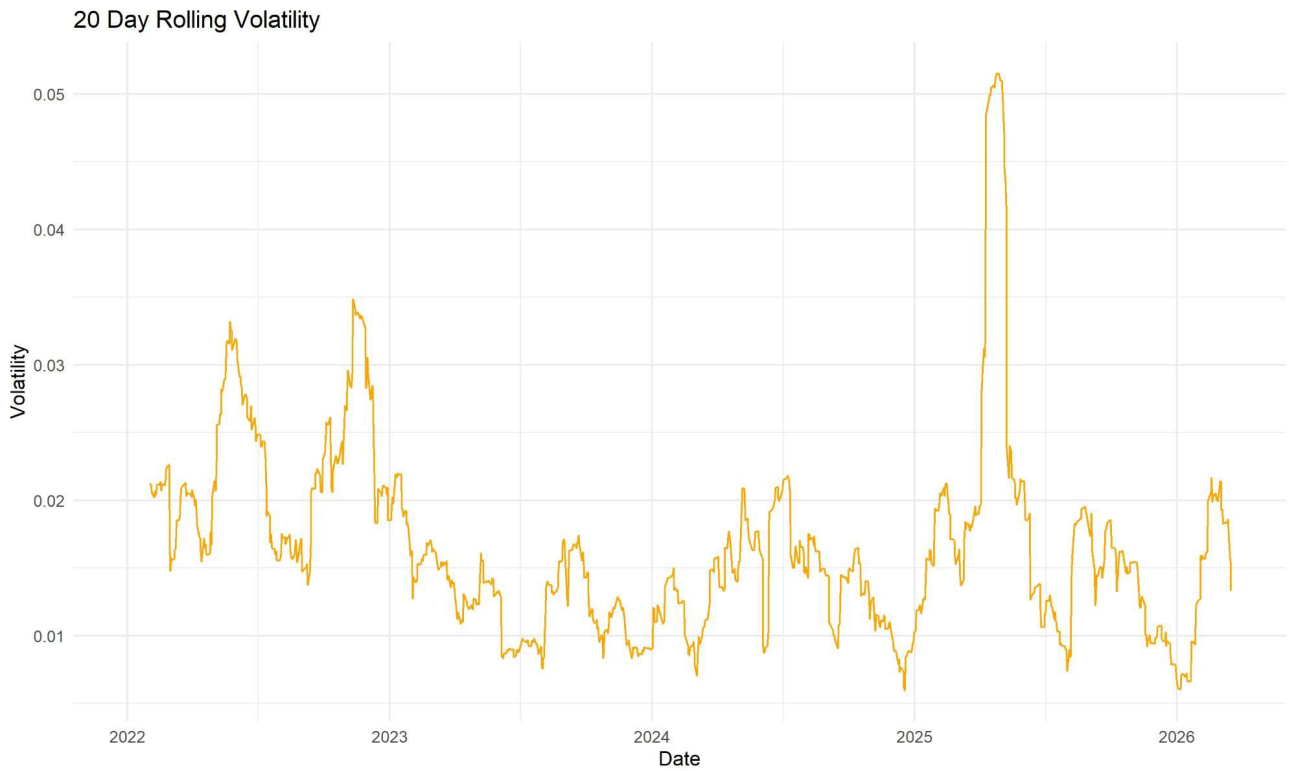

Figure 8: 20-Day Rolling Volatility

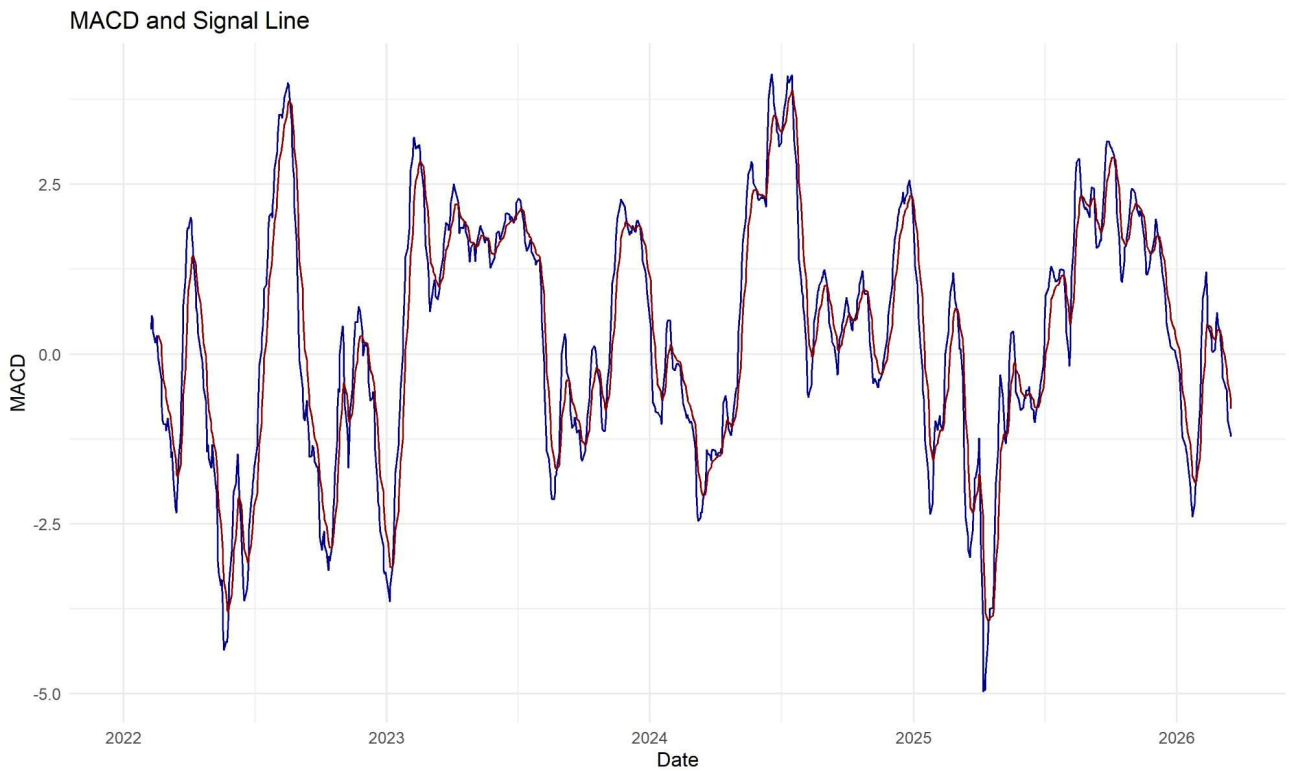

Figure 9: MACD and Signal Line

Correlation Matrix

```
          AAPL      MSFT      GOOG
AAPL  1.0000000  0.8463492  0.8986586
MSFT  0.8463492  1.0000000  0.7688684
```

```
GOOG  0.8986586  0.7688684  1.0000000
```

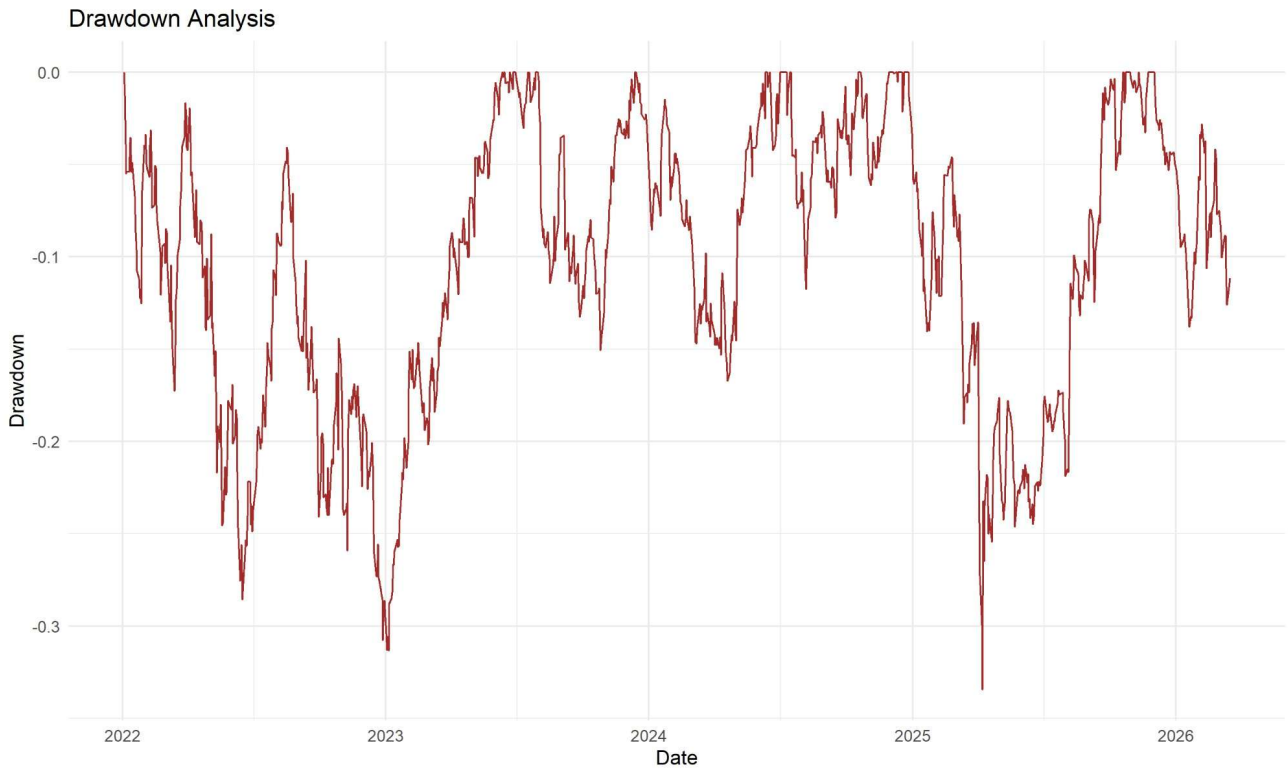

Figure 10: Drawdown Analysis

Risk Analysis

```
Standard Deviation of Returns: 0.01787168
```

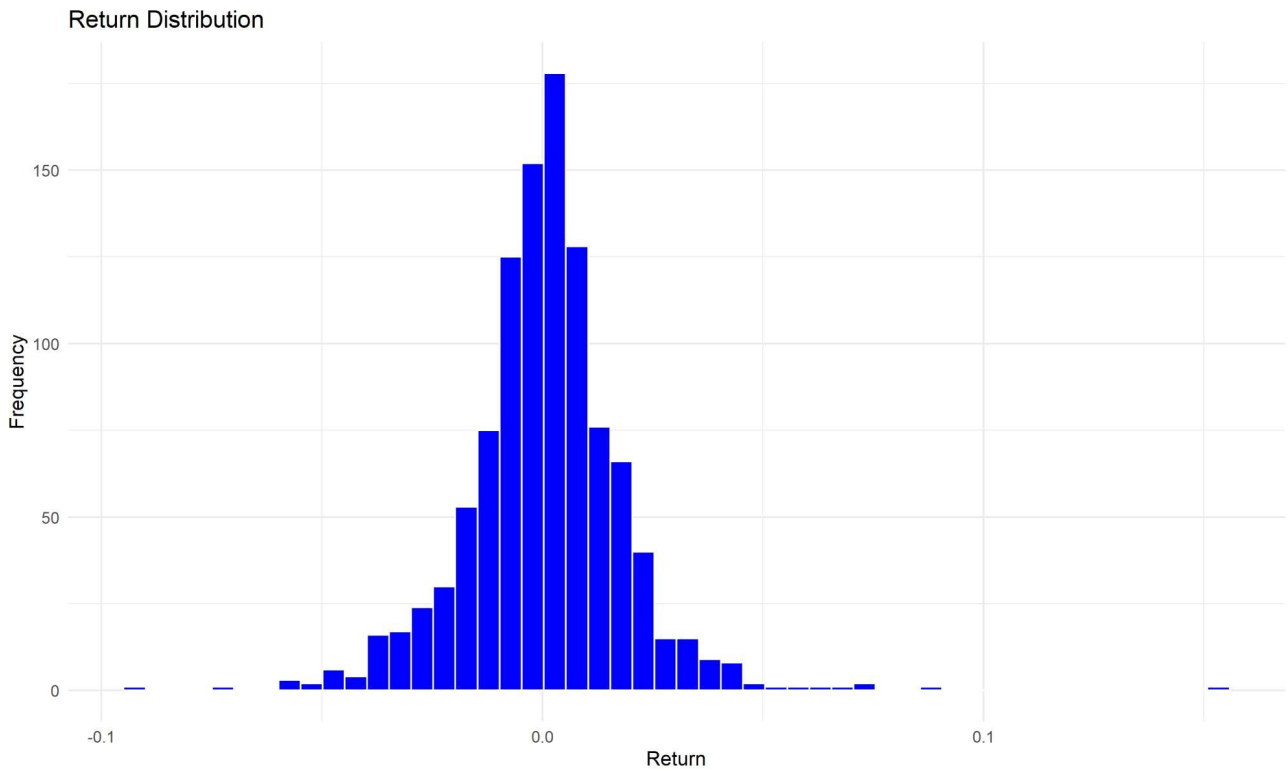

Figure 11: Return Distribution Histogram

All plots were saved successfully to the stock_plots directory and the processed stock data was exported as a CSV file. The project completed with 22 warnings, primarily related to package loading conflicts between dplyr and base R lag() function — these are standard informational warnings and do not affect output quality.

References

1. **Project Repository:** https://github.com/dheeraj-k-310/R-Library
2. **R Programming Documentation:** https://www.r-project.org
3. **ggplot2 Documentation:** https://ggplot2.tidyverse.org